What is claimed is:

1. A method for perceiving a road environment, comprising:

acquiring an associated-region lane attribute and an information to be detected, wherein the information to be detected is collected by an onboard sensor and represents a target region where a vehicle is traveling, the associated-region lane attribute corresponds to an associated region, and the associated region and the target region meet a predetermined similarity condition; and

processing the associated-region lane attribute and the information to be detected by using an onboard perception model to obtain a road perception information of the target region.

2. The method according to claim 1, wherein the onboard perception model comprises an onboard encoder and an onboard decoder, and the processing the associated-region lane attribute and the information to be detected by using an onboard perception model to obtain a road perception information of the target region comprises:

fusing a feature to be detected and an associated-region feature by using the onboard encoder to obtain a to-be-detected region feature in a bird's eye view space, wherein the feature to be detected is determined according to the information to be detected, and the associated-region feature is obtained by performing a feature extraction on the associated-region lane attribute; and

processing the associated-region feature and the to-be-detected region feature by using the onboard decoder based on an attention mechanism to obtain the road perception information.

3. The method according to claim 2, wherein the associated-region lane attribute comprises an associated-region lane position and an associated-region lane type, and the associated-region feature is determined by:

performing a spatial feature extraction on the associated-region lane attribute to obtain an associated spatial feature; and

performing a feature extraction on the associated-region lane position and the associated-region lane type to obtain an associated-lane position and type feature, wherein the associated-region feature comprises the associated spatial feature and the associated-lane

position and type feature.

4. The method according to claim 3, wherein the fusing a feature to be detected and an associated-region feature by using the onboard encoder comprises:

fusing the feature to be detected and the associated-lane spatial feature by using a first attention network of the onboard encoder to obtain a spatial fusion feature in the bird's eye view space; and

fusing the spatial fusion feature and the associated-lane position and type feature by using a second attention network of the onboard encoder to obtain the to-be-detected region feature.

5. The method according to claim 2, wherein the processing the associated-region feature and the to-be-detected region feature by using the onboard decoder based on an attention mechanism comprises:

processing a lane attribute query feature, a key feature and a value feature by using the onboard decoder based on the attention mechanism to obtain a target fusion feature, wherein the key feature and the value feature are determined based on the to-be-detected region feature and the associated-lane spatial feature, the lane attribute query feature is obtained by fine-tuning the onboard perception model, and the road perception information is determined according to the target fusion feature.

6. The method according to claim 5, wherein the onboard decoder comprises a topology classification network, and the processing a lane attribute query feature, a key feature and a value feature by using the onboard decoder based on the attention mechanism to obtain a target fusion feature comprises:

fusing the lane attribute query feature, the key feature and the value feature based on the attention mechanism to obtain an intermediate fusion feature;

processing the intermediate fusion feature using the topology classification network to obtain a classification result, wherein the classification result represents a presence or absence of a lane topology between different region lanes in the target region; and

determining the target fusion feature according to the classification result and the intermediate fusion feature based on the attention mechanism.

7. The method according to claim 6, wherein the determining the target fusion feature according to the classification result and the intermediate fusion feature based on the attention mechanism comprises:

performing a feature fusion on the intermediate fusion feature using a self-attention weight based on a self-attention mechanism to obtain the target fusion feature, wherein the self-attention weight is determined according to the classification result.

8. The method according to claim 1, wherein the road perception information comprises at least one of a region lane line, a region lane topology, and a region lane group;

wherein the region lane group represents an association relationship between at least two lanes in the target region.

9. The method according to claim 1, wherein the information to be detected is a multimodal information to be detected, and the multimodal information to be detected comprises at least one of an image to be detected and a point cloud to be detected.

10. The method according to claim 1, wherein the acquiring an information to be detected comprises:

acquiring a first information to be detected, wherein the first information to be detected is collected by an onboard sensor installed on the vehicle; and/or

acquiring a second information to be detected, wherein the second information to be detected is collected by onboard sensors installed on other vehicles different from the vehicle, and the information to be detected comprises the first information to be detected and the second information to be detected.

11. A vehicle control method, comprising:

controlling a vehicle to drive according to a road perception information, wherein the road perception information is determined according to the method of claim 1.

12. A method for training a perception model, comprising:

acquiring a sample associated-region lane attribute and a sample information to be detected, wherein the sample information to be detected represents a sample target region, the sample associated-region lane attribute corresponds to a sample associated region, and the

sample associated region and the sample target region meet a predetermined similarity condition; and

performing knowledge transfer training on an initial onboard perception model based on a knowledge distillation mechanism according to the sample associated-region lane attribute, the sample information to be detected, and a large model serving as a teacher model, so as to obtain an onboard perception model, wherein the onboard perception model is configured to process an associated-region lane attribute and an information to be detected to obtain a road perception information of a target region, and the information to be detected is collected by an onboard sensor.

13. The method according to claim 12, wherein the performing knowledge transfer training on an initial onboard perception model based on a knowledge distillation mechanism according to the sample associated-region lane attribute, the sample information to be detected, and a large model serving as a teacher model comprises:

fusing a sample feature to be detected and a sample associated-region feature using an onboard encoder of the initial onboard perception model to obtain a sample to-be-detected region feature, wherein the sample feature to be detected is determined according to the sample information to be detected, and the sample associated-region feature is obtained by performing a feature extraction on the sample associated-region lane attribute;

determining a feature loss based on the sample to-be-detected region feature and a sample collaborative to-be-detected region feature, wherein the sample collaborative to-be-detected region feature is obtained by fusing the sample information to be detected and the sample associated-region lane attribute using the large model; and

training the initial onboard perception model based on the feature loss.

14. The method according to claim 13, wherein the sample collaborative to-be-detected region feature is obtained by:

processing the sample information to be detected using a collaborative feature extraction network of the large model to obtain a sample collaborative feature to be detected;

processing the sample associated-region lane attribute using a target prompt information construction network related to the large model to obtain a sample target prompt information; and

processing the sample collaborative feature to be detected and the sample target

prompt information using an encoder of the large model to obtain the sample collaborative to-be-detected region feature.

15. The method according to claim 14, wherein the performing knowledge transfer training on an initial onboard perception model based on a knowledge distillation mechanism according to the sample associated-region lane attribute, the sample information to be detected, and a large model serving as a teacher model further comprises:

processing the sample associated-region lane attribute using an associated feature extraction network of the initial onboard perception model to obtain the sample associated-region feature; and

determining a first feature loss based on the sample associated-region feature and the sample target prompt information, wherein the feature loss comprises the first feature loss, and the feature loss further comprises a second loss between the sample to-be-detected region feature and the sample collaborative to-be-detected region feature.

16. The method according to claim 13, wherein the training the initial onboard perception model based on the feature loss comprises:

training the initial onboard perception model based on the feature loss and a road perception information loss, wherein the road perception information loss is determined based on a sample road perception information and a sample collaborative road perception information, the sample road perception information is obtained by processing the sample associated-region lane attribute and the sample information to be detected using the initial onboard perception model, the sample collaborative road perception information is configured as a pseudo-label and is obtained by processing the sample associated-region lane attribute and the sample information to be detected using the large model.

17. An electronic device, comprising:

at least one processor; and

a memory communicatively connected to the at least one processor, wherein the memory stores instructions executable by the at least one processor, and the instructions are configured to, when executed by the at least one processor, cause the at least one processor to at least:

acquire an associated-region lane attribute and an information to be detected,

wherein the information to be detected is collected by an onboard sensor and represents a target region where a vehicle is traveling, the associated-region lane attribute corresponds to an associated region, and the associated region and the target region meet a predetermined similarity condition; and

process the associated-region lane attribute and the information to be detected by using an onboard perception model to obtain a road perception information of the target region.

18. An autonomous driving vehicle, comprising the electronic device of claim 17.

19. An electronic device, comprising:

at least one processor; and

a memory communicatively connected to the at least one processor, wherein the memory stores instructions executable by the at least one processor, and the instructions are configured to, when executed by the at least one processor, cause the at least one processor to implement the method of claim 12.

20. A non-transitory computer-readable storage medium having computer instructions therein, wherein the computer instructions are configured to cause a computer to implement the method of claim 1.